

Project CAP599: Research Summary

1. Project Title

Uncovering Cross-Kingdom Heat Stress Survival Biomarkers Using a Unified Machine Learning Multi-Omics Pipeline.

2. Research Topic

The application of high-dimensional Machine Learning (Random Forest architecture) on transcriptomic (RNA-Seq) data to discover evolutionary conserved, universal thermal resilience mechanisms across both plant (*Oryza sativa*) and animal (*Bos taurus*) kingdoms.

3. Research Gap & Problem Statement

The Biological "Silo" Effect:

Historically, agricultural plant biologists and veterinary livestock researchers have operated in strict isolation. Neither side cross-references their multi-omics data to discover shared, evolutionary conserved climate resilience traits. This represents a massive, untouched research gap in universal biomarker discovery.

The Mathematical Bottleneck ($p \gg n$ Problem):

Biological RNA-Seq data contains massive noise with tens of thousands of features (genes, p) but very few samples (n). Traditional statistical methods fail in this high-dimensional space due to severe overfitting, requiring an advanced Machine Learning feature selection framework.

4. Datasets Used

The project utilizes open-source transcriptomic data from the NCBI Gene Expression Omnibus (GEO):

- **Flora Kingdom (Rice / *Oryza sativa*):** GSE153030
Link: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE153030>
- **Fauna Kingdom (Dairy Cattle / *Bos taurus*):** GSE289946
Link: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE289946>

5. Methodology

Our custom-built computational architecture executed the following step-by-step pipeline:

1. **Data Preprocessing & Noise Removal:** Transposed the raw count matrices and applied a Variance Threshold (0.1) to automatically filter out 'dead' or low-variance genes, reducing biological noise.
2. **Z-Score Normalization:** Applied StandardScaler to balance the mathematical scales across all expression values.
3. **Machine Learning Engine (Random Forest):** Deployed an ensemble of 500 decision trees utilizing the Random Subspace Method (`max_features='sqrt'`) to effectively navigate the $p \gg n$ dimensionality imbalance without overfitting. Model accuracy was validated using Out-Of-Bag (OOB) scoring.
4. **Biomarker Extraction:** Extracted the Mean Decrease in Gini Importance scores to mathematically rank the top survival genes for both species.
5. **Cross-Kingdom Intersection:** Applied an Ortholog Mapping Bridge and Set Theory algorithms to intersect the top genes and discover universally shared functional proteins.

6. Final Results

1. Kingdom-Specific Discoveries:

- **Oryza sativa:** Gene B6750 was isolated as the absolute most critical survival biomarker for rice under heat stress.
- **Bos taurus:** Gene ENSBTAG00000027477 was isolated as the absolute most critical survival biomarker for dairy cattle facing extreme heat.

2. Cross-Kingdom Universal Biomarkers:

Using the mathematical intersection, the pipeline successfully proved that both plants and animals utilize a highly conserved, shared cellular defense mechanism against extreme thermal stress. The algorithm isolated 4 Universal Biomarkers:

- **HSP70 (Heat Shock Protein 70kDa):** Acts as a biological shield preventing protein denaturation.
- **HSP90 (Heat Shock Protein 90kDa):** Aids in cellular recovery and structural integrity.
- **ROS_Scavenger (Superoxide Dismutase):** Acts as a cellular vacuum to remove toxic oxygen garbage generated by heat panic.
- **Transcription_Factor_HSF1:** The "Master Switch" that commands the DNA to deploy the heat shock shields.